1- Let R and S be two partial orders on a set A. Prove or disprove that $R \cup S$ is a partial order on A.

2- Draw the Hass diagrams of divisibility relation "$|$" on the sets $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{1, 2, 3, 6, 12, 24\}$.

Determine if $(A, |)$ or $(B, |)$ is a lattice.

3- Prove that a bipartite graph has no odd cycle.

4- Let n be an even number. Prove that the maximum number of edges of a bipartite graph wit n vertices is $\frac{n^2}{4}$.

5- Consider a graph with n verdices.

Prove that if the degree of each vertex is at least $\frac{n-1}{2}$ then the graph is connected.

Also draw a graph on $n=10$ verdices such that the degree of each vertex is $\frac{n-2}{2} = 4$ and the graph is not conneted.